Summary Notes:
AWS Fundamentals - I

1. Introduction

Overview

Here's an overview of what was covered in the session:

1. Overview of AWS
2. AWS Global Cloud Infrastructure
3. Core AWS Services
    - EC2
    - S3
    - RDS
    - VPC

2. Overview of AWS

Amazon Web Services(AWS) is the cloud computing service offered by Amazon, which was launched in 2006. Today, it offers over 200 services from data centres located globally and it is one of the most broadly adopted cloud platforms.

It is used by enterprises, startups and government agencies to power their infrastructure and applications. Organisations use AWS for lowering costs and becoming more agile.

AWS offers pay-as-you-go technology services that are delivered on-demand over the internet. Because these services are pay as you use, you may use them to create and run any kind of application without having to pay for it upfront.

It provides a range of services, ranging from emerging technologies like artificial intelligence and machine learning, data lakes and analytics, and IoT to infrastructure technologies like databases and compute storage.

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AWS has consistently accelerated technological advancement to create innovations that have the potential to completely change the industry, such as when it introduced serverless computing with the introduction of:

- AWS Lambda, developers can execute their code without having to worry about setting up or maintaining servers.
- Machine learning tools for creating, honing and implementing models are available through Amazon SageMaker, a fully managed machine learning service.

Additionally, AWS has a huge global network of AWS regions. Multiple availability zones, each physically isolated from the others but linked by low latency, high throughput, and highly redundant networks, are present in each of these regions. Because of this, it is easier to create and use programs that are:

- Highly available
- Fault tolerant
- Scalable

The AWS infrastructure meets the security standards of most risk-sensitive organisations such as the military and banks. To guarantee the availability, confidentiality and integrity of the data, it is constantly monitored. 143 security standards and compliance certifications are supported by AWS.

3. AWS Global Cloud Infrastructure

AWS Global Cloud Infrastructure is a cloud platform that is safe, scalable and dependable. It provides over 200 services from data centres all over the world. Whether it is for delivering application workloads globally in a single step or installing particular apps closer to the end users with millisecond latency, AWS offers the cloud infrastructure.

- 105 Availability Zones
- 33 Regions each with multiple Availability Zones
- 600+ CloudFront POPs and 13 Regional edge caches
- 41 Local Zones
- 29 Wavelength Zones for ultra low-latency applications
- 245 countries and territories served
- 135 Direct Connect locations

3.1 Features

3.1.1 Security

The AWS core infrastructure is constantly monitored to guarantee the availability, integrity and confidentiality of data. It is built to fulfil the strictest security standards.

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Before data leaves the secured facilities, it is automatically encrypted at the physical layer across the AWS global network.

3.1.2 Availability

Out of all cloud providers, AWS has the highest network availability. Every region consists of several Availability Zones (AZs), which are completely independent infrastructure partitions. Each region is totally isolated. Apps can be divided among several AZs in the same area to improve issue isolation and achieve high availability.

3.1.3 Performance

The performance of the AWS Global Infrastructure is its design. Low latency, low packet loss, and excellent network quality are all provided by AWS Regions. A 400 GbE fibre network backbone that is completely redundant and frequently offers several terabits of capacity between Regions is used to do this.

Because AWS Local Zones and AWS Wavelength bring AWS infrastructure and services closer to end users and 5G connected devices, they offer performance for applications that need single-digit millisecond latency.

3.1.4 Scalability

With the help of the AWS Global Infrastructure, businesses can benefit from the cloud's theoretically limitless scalability and tremendous flexibility.

With AWS, businesses can immediately scale up or down in response to their needs, which lowers costs and enhances the customer's capacity to satisfy user expectations. This allows businesses to supply only the amount of resources that they truly need.

In only a few minutes, businesses may deploy hundreds or even thousands of servers by rapidly spinning up resources as needed.

3.1.5 Flexibility

With the AWS Global Infrastructure, users can run workloads on different networks, control planes, APIs, and AWS services, giving them flexibility in how and where they execute them.

- Any AWS Region and AZ can be used to execute applications globally.
- If apps must run with single-digit millisecond latency to mobile devices and end users, there is AWS Local Zones and AWS Wavelength.
- AWS Outposts can also be used to operate apps on-premises.

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3.1.6 Global Footprint

Of all the providers, AWS has the biggest infrastructure footprint in the world, and it keeps growing at a considerable rate.

There is flexibility in choosing a technology infrastructure that is closest to the main user target when delivering workloads and applications to the cloud. The cloud can host workloads that provide optimal support for a wide range of applications, including those with the highest throughput and lowest latency needs.

4. Core AWS Services

4.1 Amazon EC2 (Elastic Compute Cloud)

One web service that offers scalable compute capacity in the AWS cloud is Amazon EC2. The initial cost of hardware is greatly decreased by using EC2.

Launching as many or as few virtual servers as necessary, configuring their networking and security, and managing storage can all be done with Amazon EC2.

When website traffic spikes, it can be scaled up to perform computation-intensive operations like monthly or annual processes. Capacity can be scaled back or decreased when usage declines.

Within the AWS Cloud, an EC2 instance is a virtual server. The hardware that the instance has access to depends on the instance that is given. Each instance type offers a different balance of compute, memory, network and storage resources.

4.1.1 Features

Amazon EC2 provides the following features:

- **Amazon Machine Images (AMIs)**
  AMIs are premade templates that AWS offers. They bundle the necessary server components, such as the operating system and extra software.

- **Types of instances**
  Different configurations for the instances' CPU, memory, storage, graphics hardware, and networking capacity.

- **Key pairs**

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Safeguard the instance login credentials. The public key is stored by AWS, while the user keeps the private key.

- **Security groups**
  A virtual firewall that lets you set the ports, source IP ranges, IP addresses and protocols that the instances can connect to and receive traffic from.

4.1.2 Use cases

- Run cloud-native and enterprise applications.
- Scale for High Performance Computing (HPC): Access to on-demand infrastructure and capacity needed to run large, complex simulations and deep learning workloads faster and cost-effectively.
- Develop Apple applications: AWS provides EC2 Mac instances where one can develop, build and test on-demand macOS workloads.
- Train and deploy ML applications: Provides high-bandwidth network (up to 400 Gbps), huge processing compute and storage services purpose built for high performance ML applications

4.1.3 Examples

- Netflix needs to scale very quickly as it has temporarily higher viewing and increased member growth.
- Edmodo has been able to quickly scale with EC2 instances as the site capacity increased by 15 times, resulting in uninterrupted services for the teachers and students.

4.2 Amazon S3 (Simple Storage Service)

AWS Simple Storage Service or simply S3 is one of AWS's core services which allows users to store objects in the cloud and can be used for a large range of things from large scale data lakes to static websites.

It provides management features to organise, optimise and configure access to the data to meet specific organisational and compliance requirements.

S3 can virtually hold an unlimited amount of data and is a cost effective storage solution.

4.2.1 How does it work ?

Data is stored as objects within buckets using Amazon S3, an object storage service. A file and any metadata that pertains to it are considered objects. An object's container is a bucket.

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Access rights must be explicitly granted in order to access buckets and their contents, which are private. To regulate access, you can use S3 Access Points, AWS Identity and Access Management (IAM) rules, bucket policies and access control lists (ACLs).

4.2.2 Features

- **Storage classes**
  S3 provides a variety of storage classes designed to meet different use cases. For example, for frequent access there's S3 Express One Zone, or S3 Standard for critical production data, or S3 Standard-IA for storing infrequently accessed data and saving costs. There's also S3 Glacier for archive data.

- **Storage management**
  S3 has features that can be used to reduce latency, save multiple copies of the data and manage costs. Objects can be transitioned to other S3 storage classes. There's Object Lock which prevents objects from being overwritten or deleted for an indefinite or fixed amount of time. To reduce latency, objects can be replicated to many destination buckets in the same or different AWS regions. Batch Operations provide management of billions of objects at scale.

- **Access management and security**
  Access control and auditing to buckets and objects are provided by Amazon S3. By default, S3 objects and buckets are private, but S3 Block Public Access allows you to restrict public access to them. AWS Identity and Access Management (IAM) provides secure access control to AWS resources.

- **Data processing**
  The ability to add custom code to S3 GET, HEAD, and LIST queries allows users to alter and process data as it is returned to an application using S3 Object Lambda. It enables row filtering, redacting sensitive data and dynamic image scaling. Event notifications are sent out via Amazon Simple Notification Service (Amazon SNS) and Amazon Simple Queue Service (Amazon SQS) in response to modifications made to S3 resources.

4.2.3 Use Cases

- Data lake can be built with big data analytics, ML, AI and high performance computing (HPC) applications.
- Backup and restore data with S3 robust replication features.
- With Amazon S3 Glacier storage class, archive data at lower costs.
- Cloud native applications which scale automatically can be built and run.

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4.2.4 Examples

- Shutterstock catalogue has more than 405 million images and 25 million videos. Each day it increases by approx 200,000 images
- Snap (Snapchat) stores 2 exabytes which is over 1.5 trillion photos and videos
- NASCAR library grows at roughly 1.5 PB to 2 PB of data per year

4.3 Amazon RDS (Relational Database Service)

Within the AWS Cloud, Amazon RDS is a web service for relational databases. It facilitates simple setup, operation, and demand-based scaling. It offers reasonably priced, scalable capacity for well-known open-source and proprietary relational database engines, including:

- Amazon Aurora which is a MySQL and PostgreSQL compatible database built for the cloud by AWS.
- Db2
- Oracle
- MySQL
- MariaDB
- PostgreSQL
- Microsoft SQL Server

Amazon automates the undifferentiated management task for the database such as provision, configuration, backup and patching.

4.3.1 Features

- **Easy management**
  A new database instance can be created and connected with the application within minutes, as Amazon RDS database instances are preconfigured with settings and parameters appropriate for the selected engine and class.

- **Rich engine choices**
  Deploy and scale the relational database engines from the various industry-standard engines such as Db2, MariaDb, Microsoft SQL Server, MySQl, Oracle, PostgreSQL and Amazon Aurora.

- **High availability and durability**
  Amazon RDS Multi-AZ offers enhanced availability and durability of database instances as when it is provisioned, Amazon RDS synchronously replicates the data to a standby instance in a different AZ. Amazon RDS provides automated backup features which enables point-in-time recovery of the database instance.

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- **High scalability**
  Scale the compute and memory resources powering the deployment up and down in a few minutes, up to a max of 128 vCPUs and 4096 GiB of RAM. Amazon Aurora automatically increases the size of the database volume as database storage needs grow.

4.3.2 Use cases

- Build mobile and web applications with high throughput, availability and storage scalability.

4.3.3 Examples

- Inuit Mint uses Amazon RDS for MySQL which helps them provide security for the 50 terabytes of financial data it stores.
- Cathay Pacific moved its system to the cloud using Amazon RDS for Oracle as its passenger revenue optimisation system (PROS) on premises incurred a heavy maintenance burden on their suboptimal compute power and small DevOps team.
- Samsung migrated 1.1 billion users from Oracle to Amazon Aurora with AWS Database MigrationService.

4.4 Amazon VPC (Virtual Private Cloud)

You can launch AWS resources in a virtual network of your choosing by establishing a logically isolated area of the AWS cloud using Amazon VPC.

By using Amazon VPC, users may customise the network architecture and manage how instances connect to each other, the internet, and other networks. This involves setting up network gateways, route tables, and subnets in addition to choosing an IP address range.

You can link your corporate datacenter to the VPC with a hardware Virtual Private Network (VPN).

Access to Amazon EC2 instances within each subnet is managed by a number of security layers, such as security groups and network access control lists.

4.4.1 Features

- **Subnets**
  The resources in a VPC can be grouped according to security and operational requirements by creating subnets. Subnets may be private or public.

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- **Route tables**
  Traffic routing between subnets and between your VPC and external networks is managed using route tables. Specific routing rules can be defined via custom route tables.

- **Security groups**
  Security groups regulate incoming and outgoing traffic for your instances by acting as virtual firewalls.

- **VPN and Direct Connect**
  Use a VPN connection to establish a secure link between a VPC and an on-premises network. A dedicated network connection from your location to AWS is offered by AWS Direct Connect.

- **Increased security**
  Secure and monitor connections, restrict access of instances inside the virtual network and screen traffic.

- **Ease of management**
  Customise virtual networks by choosing IP address ranges, configuring route tables, and creating subnets.

4.4.2 Use cases

- Deploy a web application in a public subnet, with backend servers and databases in private subnets. Use security groups and network ACLs to control access.
- Implement a multi-tier architecture with separate subnets for web servers, application servers and databases. Use routing rules to control communication between tiers.
- Improve the security of your web application by enforcing rules on inbound and outbound connections.
- Define network connectivity and restrictions between databases, application servers, and web servers.

In this session, we covered an overview of AWS and its core services. AWS offers a comprehensive and reliable cloud platform that supports a wide range of applications with its vast global infrastructure.

Understanding AWS services like EC2, S3, RDS and VPC, along with the key features and use cases, provides a solid foundation of AWS. AWS's focus on security, scalability and flexibility makes it a powerful tool for any organisation looking to innovate and optimise its operations in the cloud.

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